BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz 3 | Semester: Summer 2025
Duration: 25 minutes | Full Marks: 20

CSE 340: Computer Architecture

Name:	ID:	Section:

1.

a) Study the given RISC-V Datapath thoroughly. Assume that initially **PC=3FAh, x1 = 103h, x2 = EAAh, x3 = CF4h, x4=999h**. Determine the values of A, B, C, D, E in hex for the instructions executed sequentially. If the value does not matter, write it as X (don't care). — 5

		A	B	C	D	E
i	`SUB X2, X3, X4`					
ii	`sd x2,0(X0)`					

b) Determine the values of the following control bits when executing the instruction "`And X21, X22, X23`" in the provided single-cycle datapath. — 5

Memread =	Branch =	ALUSrc =	RegWrite =	MemToReg =

2. The following RISC-V instructions are running on a pipelined datapath. Answer the following questions based on this.

```
add X1, X2, X7
sub X2, X1, X5
ld X2, 16(X5)
sd X2, 16(X1)
sll X5, X6, X7
ld X2, 16(X5)
add X2, X1, X7
```

a) Determine the number of hazards present in the code.[1]

Answer:

b) How many clock cycles are needed to run the instruction set? [1]

Answer:

Instruction Fetch takes 15ps, Register read/write takes 15ps, ALU operation takes 30 ps, Memory access takes 15ps.

c) Find the clock period. [1]

Answer:

d) Total time to run the instructions. [1]

Answer:

e) Solve the data hazards using only stalling. [5]

Answer:

f) After resolving the hazards, by how much did the execution time change?[1]

Answer:

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1.	a) Study the given RISC-V Datapath thoroughly. Assume that initially **PC=FA3h, x1 = 183h, x2 = E0Ah, x3 = CF3h, x4=991h**. Determine the values of A, B, C, D, E in hex for the instructions executed sequentially. If the value does not matter, write it as X (don't care)	5

		A	B	C	D	E
i	`ADD X2, X3, X4`					
ii	`AND X2, X1, X4`					

	b) Determine the values of the following control bits when executing the instruction "`BEQ X1, X6, FUNC`" in the provided single-cycle datapath.	5

Memread =	Branch =	ALUSrc =	RegWrite =	MemToReg =

2. The following RISC-V instructions are running on a pipelined datapath. Answer the following questions based on this.

```
sub X11, X21, X22
and X21, X11, X14
lw X21, 16(X14)
sw X21, 16(X11)
srli X14, X6, 22
ld X21, 16(X14)
addi X21, X11, 22
```

a) Determine the number of hazards present in the code.[1]

Answer:

b) How many clock cycles are needed to run the instruction set? [1]

Answer:

Instruction Fetch takes 25ps, Register read/write takes 20ps, ALU operation takes 40 ps, Memory access takes 25ps.

c) Find the clock period. [1]

Answer:

d) Total time to run the instructions. [1]

Answer:

e) Solve the data hazards using only stalling. [5]

Answer:

f) After resolving the hazards, by how much did the execution time change?[1]
Answer:

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1.

a) Study the given RISC-V Datapath thoroughly. Assume that initially **PC=63Ah, x1 = 400h, x2 = ABCh, x3 = 0F4h, x4=444h**. Determine the values of A, B, C, D, E in hex for the instructions executed sequentially. If the value does not matter, write it as X (don't care) [5]

		A	B	C	D	E
i	`ADD X2, X3, X4`					
ii	`SLLI X3, X2, 4`					

b) Determine the values of the following control bits when executing the instruction "`LD X9,64(X10)`" in the provided single-cycle datapath. [5]

Memread =	Branch =	ALUSrc =	RegWrite =	MemToReg =

2. The following RISC-V instructions are running on a pipelined datapath. Answer the following questions based on this.

```
add X4, X5, X10
sub X5, X4, X8
lh X5, 64(X8)
sh X5, 32(X4)
sll X8, X9, X10
ld X5, 16(X8)
add X5, X4, X10
```

a) Determine the number of hazards present in the code.[1]

Answer:

b) How many clock cycles are needed to run the instruction set? [1]

Answer:

Instruction Fetch takes 35ps, Register read/write takes 25ps, ALU operation takes 50 ps, Memory access takes 35ps.

c) Find the clock period. [1]

Answer:

d) Total time to run the instructions. [1]

Answer:

e) Solve the data hazards using only stalling. [5]

Answer:

f) After resolving the hazards, by how much did the execution time change?[1]
Answer:

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1.	a) Study the given RISC-V Datapath thoroughly. Assume that initially **PC=342h, x1 = 200h, x2 = 148h, x3 = 1F2h, x4=456h**. Determine the values of A, B, C, D, E in hex for the instructions executed sequentially. If the value does not matter, write it as X (don't care)	5

		A	B	C	D	E
i	`ADD X2, X3, X4`					
ii	`SD X3, 16(X2)`					

b) Determine the values of the following control bits when executing the instruction "`SD X9,64(X10)`" in the provided single-cycle datapath. 5

Memread =	Branch =	ALUSrc =	RegWrite =	MemToReg =

2. The following RISC-V instructions are running on a pipelined datapath. Answer the following questions based on this.

```
add X12, X13, X18
sub X13, X12, X16
lh X13, 64(X16)
sh X13, 32(X12)
sll X16, X17, X18
ld X13, 16(X16)
add X13, X12, X18
```

a) Determine the number of hazards present in the code.[1]

Answer:

b) How many clock cycles are needed to run the instruction set? [1]

Answer:

Instruction Fetch takes 20ps, Register read/write takes 15ps, ALU operation takes 30 ps, Memory access takes 20ps.

c) Find the clock period. [1]

Answer:

d) Total time to run the instructions. [1]

Answer:

e) Solve the data hazards using only stalling. [5]

Answer:

f) After resolving the hazards, by how much did the execution time change? [1]

Answer: